

William Lu

(832) 682-6054 chenjingkun65@gmail.com linkedin.com/in/william-lu-0x57-0x4c william-f-12.github.io

Education

University of Illinois Urbana-Champaign Sep 2024 - Expected May 2027
B.S. in Mathematics & Computer Science GPA: 4.0/4.0
Previous study: University of Wisconsin–Madison, Computer Science (2023–2024) GPA: 4.0/4.0
Relevant Coursework: Machine Learning, Applied Parallel Programming, Algorithms, Computer Systems
Languages: Mandarin (Native), English (Fluent), Japanese (Functional)
Technical Skills: C/C++, Python, Java, C#, JavaScript; PyTorch, CUDA, Django, Unity, LangGraph; Perfetto, Nsight Systems, Nsight Compute, ADB, RISC-V, Git
Publication: Comparisons of LSTM and other Machine Learning Approaches on Predicting Suicidal Tendency Regarding to Social Media Posts (2021)

Experience & Projects

TikTok - Effect Core SDK San Jose, CA
Software Engineer Intern May 2026 – Aug 2026

- Built an auto-trace collection-to-report platform (Python, Perfetto, ADB/UIAutomator), replacing a 10-step manual workflow with a single command and producing a 1,200-trace, 482-effect corpus across 6 device profiles; let two other engineers onboard new hardware and extend the collector without touching core logic.
- Designed the multi-layered agentic performance attribution and prioritization analysis tool behind it, giving engineers a ranked list of which stage would yield the highest overall improvement if optimized. This project provided the data foundation for the Effects Performance Team's Q3 optimization roadmap.
- Established end-to-end performance instrumentation from high-level SDK down to individual JS functions in the effect package so engineers now get a complete picture inside a single Perfetto trace.

Convolutional Layer CUDA Kernel Optimization Champaign, IL
CUDA Developer Jan 2026 – May 2026

- Implemented and optimized convolution as CUDA matrix multiplication using Tensor Cores, shared-memory tiling, constant-memory caching, loop unrolling, CUDA streams, and mixed-precision computation, reducing combined convolution kernel time from 74.9 ms to 53 ms (29% reduction, $1.41\times$ speedup) at batch size 10,000.
- Profiled GPU execution with Nsight Compute and Nsight Systems to analyze memory traffic, instruction mix, Tensor Core utilization, and bottlenecks; Tensor Core optimization alone reduced second-layer kernel time by 21%.

Job Application Management Web App Champaign, IL
Backend Engineer Sep 2025 – Dec 2025

- Built Django REST APIs and relational data models for a team-developed job application tracking platform, supporting CRUD operations across users, applications, companies, and related resources.
- Designed normalized SQLite schemas with foreign-key constraints and serializer validation to maintain relational integrity.

Self-Supervised Representation Learning with SimCLR (PyTorch) Champaign, IL
Independent Developer Aug 2025 – Oct 2025

- Reproduced the SimCLR self-supervised learning pipeline in PyTorch based on the original paper, implementing data augmentation, a ResNet encoder, projection head, NT-Xent contrastive loss, and downstream evaluation.
- Investigated training scalability and reproduction gaps under limited compute, experimenting with batch-size constraints, gradient accumulation, and efficient data loading while training on Google Colab.

Tiny-Chat Champaign, IL
Programmer & Assistant Aug 2025 – Oct 2025

- Contributed to a multi-agent conversational framework that orchestrates LLM-powered agents to simulate social interactions including negotiation, persuasion, and collaboration.
- Implemented environment initialization and agent-profile loading from local and Hugging Face datasets, along with structured conversation logging and JSON-based persistence for experiment records.

Unix-like Kernel Champaign, IL
Programmer & designer May 2025 – Aug 2025

- Led a 3-person team in building a Unix-like kernel in C for RISC-V, implementing multitasking, virtual memory, a file system, system calls, and device drivers.
- Implemented a 2-way set-associative cache, process management, Unix pipes for IPC, and block/entropy device drivers.